Analysis of different CNN architectures for tomato grading

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***Abstract*—This paper shows an analysis of four Convolutional Neural Network (CNN) architectures for binary grading of table tomatoes as "good" or "bad", with an emphasis on defect detection. The evaluated models were InceptionV3, ResNet-50, Xception, and MobileNetV3 Large, and three datasets were considered: one publicly available dataset collected in a controlled environment, one acquired by the authors primarily in an industrial setting, and a merged dataset combining both. All architectures were evaluated on three datasets using accuracy, recall, precision, and F1-score. The ResNet-50 architecture proved to be the most promising for the problem, as it struck a good balance between recall and accuracy across the three evaluated datasets, achieving a recall ranging from 89% to 97%. The results indicate that CNN-based approaches are effective for assessing tomato quality in industrial settings, supporting the continued development of a deep learning-based perception module for an automated robotic tomato-sorting system for Brazilian industry.**

***Index Terms*—convolutional neural networks, transfer learning, tomato grading, image classification, agricultural automation, deep learning**

I. Introduction

Brazil is the world's 9th-largest tomato producer according to the Food and Agriculture Organization of the United Nations (FAO) [1], and Goias state leads national production, followed by Minas Gerais in 3rd place, with Araguari city leading Minas Gerais' production [2]. In this large-scale production scenario, the post-harvest stage presents many challenges, such as classifying fruits with defects. This grading considers many variables, such as the fruit's irregular shape, color, adhesion, and surface irregularities, as well as foreign bodies encountered during classification [3].

In this way, the quality of the product depends on the methods used [4], and this can influence consumers' buying decisions [5]. Currently, the main approach to grading is still done manually, which is vulnerable to biases and limits the scale of fruit production [6] [7]. Automating this stage requires a robotic sorting system with a machine-vision perception module and a mechanical actuation stage, such as a robotic arm that removes defective fruits from the line.

In this scenario, in Brazil, to incentivize national competitiveness, the Brazilian Ministry of Finance created the "New Industry Brazil" industrial public policy [8], which seeks to boost the development of national industry through, among other measures, investment in the development of national machinery and equipment to increase the competitiveness of Brazilian agribusiness in a global scenario and limit external dependencies for the various stages of the final production.

The benefits of applying deep learning in agriculture were demonstrated in Kamilaris and Prenafeta-Boldú [9], showing their application to many challenges, such as weed identification, land cover classification, plant recognition, fruit counting, and crop type classification. Different from traditional machine learning approaches, deep learning brings a lot of advantages, as they allow large-scale applications and its use in automation lines because it removes the need for manual feature extraction, a difficult and expensive process highly dependent on experts, which also demands revisions when the dataset changes [9]. This advantage stems from deep learning models' ability to automatically learn relevant features from data, enabling the identification of a wide range of object characteristics.

In this scenario, this work aims to demonstrate the viability of Convolutional Neural Networks (CNNs) as the perception module of a robotic sorting cell for table tomatoes, evaluating the classification of fruits into two categories ("good" or "bad") labeled by specialists.

The resulting classifier is designed for future integration into a robotic arm or conveyor-based rejection mechanism for a Brazilian industrial tomato-grading cell. This development is aligned with the mission "sustainable and digital agroindustrial chains" of "New Industry Brazil" which defines the goal of "increase the share of agroindustry in agricultural GDP to 50%, achieve 70% mechanization in family farming and ensure that at least 95% of the machinery and equipment used is

manufactured domestically" [8].

II. Related work

As mentioned in Section I, CNNs have been applied to a wide range of agricultural challenges, including fruit classification for ripeness, defects, and diseases [10] [11]. Convolutional Neural Networks are also used for tomato grading, including such classes as defect, ripeness, and size [5] [6] [7] [12].

In Amna et al. [10], CNNs were applied to defect detection on mangoes and tomatoes, achieving detection accuracies of 89% and 92% for mangoes and tomatoes, respectively. Ashtiani et al. [11] used several CNN architectures (DenseNet, InceptionV3, ResNet-18, ResNet-50, and AlexNet) to classify mulberries by ripeness stage. Among these, AlexNet and ResNet-18 delivered the best results, achieving overall accuracies of 98.32% and 98.65% for determining the ripeness of white and black mulberries, respectively. Furthermore, ResNet-18 was also able to simultaneously classify mulberry genotype and ripeness with an overall accuracy of 98.03%, surpassing the performance of the other evaluated architectures.

In tomato grading, CNNs also show strong potential. Shuaeb et al. [5] achieved 99.1% and 98.6% accuracy using ResNet-50 and AlexNet, respectively, in a four-class classification problem. Mputu et al. [6] used a hybrid CNN-SVM approach and reached an accuracy of 97.50% in the binary classification ("healthy" or "rejected") using the InceptionV3 architecture as a feature extractor. The proposed architecture also achieved 97.54% accuracy when deployed on a public dataset to categorize tomatoes as ripe, unripe, old, or damaged. Eloge [7] obtained a 98% accuracy in four-class classification (ripe, unripe, damaged, and uneven ripe) using a ResNet-50 Vision Transformer hybrid architecture. Chaturvedi et al. [12] obtained an accuracy of 97.97% in the binary classification ("healthy" or "defective") using the EfficientNetB4 architecture.

Unlike the studies discussed above (none of which evaluate a classifier trained with data collected directly on an uncontrolled industrial line), our work introduces an original industrial dataset acquired under real lighting conditions and assesses how combining it with a controlled public dataset affects the generalization capability of four CNN architectures, a necessary step toward validating a vision module for real robotic deployment. Table I shows a summary of the literature on deep learning-based methods for assessing fruits.

This study compared three different datasets in four benchmark architectures, aiming to understand the impact of image quantity on the tomato grading problem as well as to develop a methodological approach to validate the future development of a Brazilian tomato-grading technology aligned with the "New Industry Brazil" [8]. The selected architectures were InceptionV3, ResNet-50, Xception, and MobileNetV3 Large. The relevance of each one is discussed in Subsection III-B.

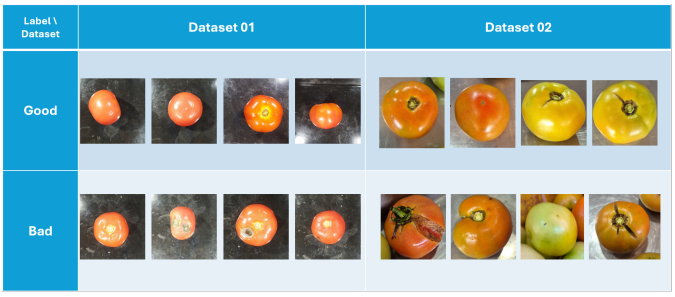


Fig. 1. Samples from Datasets 01 and 02.

III. Proposed approach

A. Dataset description

As mentioned in Section I, this work uses three tomato datasets, as described in Table II.

Dataset 01 was publicly available in Rahaman and Sarkar [13] and previously labeled by its authors, and contains 1,000 tomato images collected in a controlled environment. Dataset 02 was collected by the authors using a Samsung S24+ smartphone camera at a tomato processing plant in the interior of Minas Gerais, Brazil, under uncontrolled daylight conditions. Some samples from the "good" category were collected from tomatoes from local stores for class balancing. Specialists labeled all the samples. To understand the impact of the number of images and types of defects on the performance of the architectures, Dataset 03 is a merge of Datasets 01 and 02. Samples of the data from Datasets 01 and 02 are shown in Fig. 1. This work focuses its experiments on binary classification ("good" and "bad") due to the major challenge faced by the local business unit where the authors collected the samples.

B. Convolutional Neural Networks

This work aimed to compare the performance of four different CNN architectures using a transfer-learning approach with ImageNet weights, each selected for its relevance to related work. The rationale for selecting each of these architectures (InceptionV3, ResNet-50, Xception, and MobileNetV3 Large), and their relevance to prior fruit-grading literature and their distinct architectural trade-offs, is detailed below.

The Inception architecture [16] was selected for its relevance in the related literature and its frequent use as a benchmark [6] [11] [12]. The model was developed to scale neural networks with a focus on computational efficiency, maintaining a low number of parameters through convolution factorization: instead of using large spatial convolutions, they are factored into smaller ones. The version of Inception used in this study was InceptionV3.

The ResNet architecture introduces a residual learning structure that facilitates training of deep neural networks, enabling greater depth than previous networks (e.g., VGG) while maintaining lower complexity [17], and was used in related studies about tomato grading [7] [12]. ResNet uses shortcut connections (*skip connections*), allowing layers to learn just

TABLE I
LITERATURE REVIEW ON CNN USED FOR TOMATO GRADING.

Area of application	Method	Results	Study
Mango and tomato fruit quality detection	CNN	Detection accuracy for mango and tomato was 89% and 92%, respectively	[10]
Mulberry Ripeness Stages	DenseNet, InceptionV3, ResNet-18, ResNet-50, and AlexNet	AlexNet and ResNet-18 achieved overall accuracies of 98.32% and 98.65% for determining the ripeness of white and black mulberries, respectively. ResNet-18 also simultaneously classified mulberry genotype and ripeness, achieving an overall accuracy of 98.03%.	[11]
Tomato grading	ResNet-50 and AlexNet	Accuracy of 99.1% and 98.6% using ResNet-50 and AlexNet respectively, in four-class classification problem	[5]
Tomato grading	CNN-SVM hybrid architecture	Accuracy of 97.50% in the binary classification ("healthy" or "rejected")	[6]
Tomato grading	ResNet-50-Vision Transformer hybrid architecture	Accuracy rate of 98%, in four-class classification problem	[7]
Tomato grading	VGG19, ResNet50, DenseNet201, EfficientNetB4, and Inceptionv3	Accuracy of 97.97% in the binary classification ("healthy" or "defective") using EfficientNetB4 architecture	[12]

TABLE II
DATASETS USED IN THIS STUDY.

Dataset	Number of Images	"Good"	"Bad"
Dataset 01 [13]	1000	500	500
Dataset 02 (authors [14])	349	168	181
Dataset 03 (merging Datasets 1 and 2)	1349	668	681

the residual from previous layers. From this perspective, a deep residual network can be understood as a stack of residual units (RUs), each consisting of a small neural network with a skip connection [15]. This work uses a ResNet-50.

The Xception architecture [18], on the other hand, builds on the Inception architecture, uses depthwise separable convolutions, and a feature extraction base with 36 convolutional layers. Xception replaces the Inception modules with separable convolutional layers, which consist of two parts: "the first part applies a single spatial filter for each input feature map, then the second part looks exclusively for cross-channel patterns" [15]. This architecture demonstrated considerably superior performance to InceptionV3 in a challenge involving 350 million images and 17,000 classes [15].

On the other hand, the architecture of MobileNets presents computationally efficient models built primarily using depthwise separable convolutions, enabling the development of systems for mobile and embedded devices [19]. This computational efficiency can be a significant advantage for the proposed problem, given its need for industrial applications, where computational cost is a key consideration. In this study, MobileNetV3 Large was used because its architecture balances computational efficiency and performance [20].

We chose Xception because it is a derivative of Inception and has shown superior results in the literature [15], while MobileNet enables evaluation of neural network performance on potential embedded systems.

IV. EXPERIMENTAL SETUP

For these experiments, CNNs were developed using Python (3.14) and the Keras API, along with the following libraries: Pandas (2.2.2), NumPy (2.0.2), Matplotlib (3.10), Seaborn (0.13.2), TensorFlow (2.20), and Scikit-Learn (1.6.1). Model architecture selection, hyperparameter configuration, and the train/validation/test splitting logic were manually specified by the authors. Code and datasets are available in [14]. The T4 environment from Google Colab was used due to local environment limitations, providing 12.7GB of RAM and 15GB of GPU memory.

A. Training phase

The process used from data acquisition to evaluation is represented in Fig. 2. The samples from Dataset 01 were cropped to remove borders that could harm model training, focusing on the classification object. The samples for Dataset 02 were mostly collected in the field, with a small number of images from the "good" category collected from the local market for class balancing, and subsequently classified by experts. Finally, Dataset 03 was constructed by merging the other datasets.

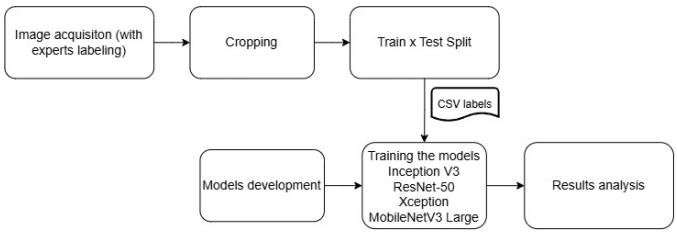


Fig. 2. Experiments workflow.

Subsequently, each dataset was split into training and test folders using an 80% training, 20% test ratio, randomly while maintaining class balance, using the Scikit-Learn library and a random seed of 42. A .csv file was generated containing the image names and labels for each dataset to ensure reproducibility. The files and code are available in the author's GitHub repository [14]. During training, 20% of the training

set data was used for validation, following a stratified hold-out strategy. The hyperparameters were set as in Table III.

To avoid overfitting, data-augmentation techniques were used with the following parameters: “rotation_range=20”, “width_shift_range=0.2”, “height_shift_range=0.2”, “horizontal_flip=True”, “zoom_range=0.2”.

The same optimizer and learning rate were used across all four architectures to isolate the effect of the backbone itself on classification performance, following the transfer-learning protocol recommended; a per-architecture hyperparameter search was not performed and is left as future work.

The images were resized to the default dimensions for each model’s input (height (H) x Width (W) x Channels (C)) as described in Table IV.

TABLE III
MODEL TRAINING HYPERPARAMETERS AND CONFIGURATIONS

Hyperparameter	**Value / Setting**	**Purpose / Note**
Optimizer	Adam	*Learning rate* = 0.001 [15]
Maximum Epochs	200	Total training cycles
Batch Size	32	Samples per gradient update
Early Stopping	*Patience* = 5	To prevent model overfitting

TABLE IV
DEFAULT INPUT DIMENSIONS FOR THE EVALUATED CNN ARCHITECTURES

Model Architecture	**Default Input Shape (H $\times$ W $\times$ C)**
InceptionV3	$299 \times 299 \times 3$
ResNet-50	$224 \times 224 \times 3$
Xception	$299 \times 299 \times 3$
MobileNetV3Large	$224 \times 224 \times 3$

B. Evaluation metrics

To evaluate the performance of each architecture, accuracy is the most commonly reported metric in the literature [9]. However, considering this specific challenge, it’s necessary to consider the risk caused by false positives and false negatives, because they can cause food waste or affect the quality of the product available in the market [6] [12], impacting consumers’ perceptions [5] [6], respectively. In this way, it’s also necessary to calculate recall and precision. Given the importance of balancing precision and recall, the F1-score is also a relevant metric.

V. RESULTS AND DISCUSSION

A. Overall performance

Each of the four architectures exhibited varying performance across the three datasets, revealing distinct aspects of its generalization capacity in different scenarios. All four architectures performed better on Dataset 02, as shown in Table V, suggesting that Dataset 02 may be an easier challenge for the networks because the classes are easily distinguishable, as shown in Fig. 1.

In other words, the results on Dataset 01 indicate a more challenging problem, with greater complexity in distinguishing between classes. One of the challenges that may have been noticed by the models is the occurrence of specular shine, due to the characteristics of the fruit and the collection environment, as shown in Fig. 1, suggesting the necessity of training the model with more images and more fine-tuning, or the need to control the environment light for effective classification of this kind of defect. The results on Dataset 03, using the defined metrics, indicate improvement over those on Dataset 01, suggesting an increase in the model’s capacity to generalize when exposed to a wider range of samples and defect types, despite lower accuracy than that obtained on Dataset 02. The accuracy and loss curves in Figs. 3 to 5 reinforce the differences between the datasets, showing faster and more stable convergence in Dataset 02 and a more irregular behavior in Dataset 01, confirming the more complex decision space.

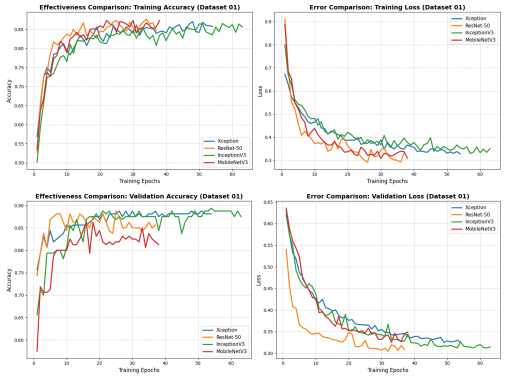


Fig. 3. Comparative learning curves (training and validation accuracy and loss) of the evaluated CNN architectures on Dataset 01.

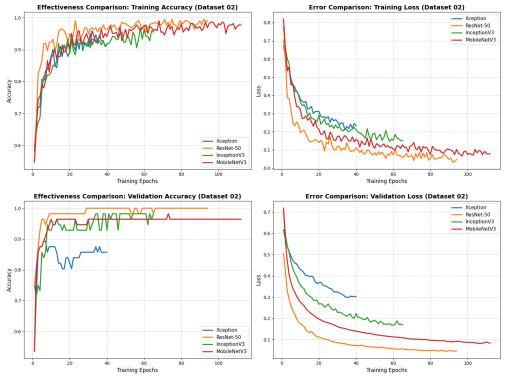


Fig. 4. Comparative learning curves (training and validation accuracy and loss) of the evaluated CNN architectures on Dataset 02.

B. Analysis by model

As mentioned in Table III, early stopping was used with a patience of 5. This way, it was expected that the model would stop after 5 consecutive epochs without learning. This behavior can be noticed in Figs. 3 to 5.

The InceptionV3 architecture achieved the best overall accuracy on Dataset 02 (97.14%), which presents clearer differences between the classes, as shown in Subsection V-A,

TABLE V
PERFORMANCE OF THE EVALUATED CNN ARCHITECTURES ACROSS THE THREE DATASETS

CNN	Performance	Dataset 01	Dataset 02	Dataset 03
InceptionV3	Accuracy	86.50%	97.14%	88.89%
	Recall	83.00%	97.22%	91.91%
	Precision	89.25%	97.22%	86.81%
	F1-Score	86.01%	97.22%	89.29%
ResNet-50	Accuracy	87.50%	94.29%	88.52%
	Recall	89.00%	97.22%	92.65%
	Precision	86.41%	92.11%	85.71%
	F1-Score	87.68%	94.59%	89.05%
Xception	Accuracy	88.00%	94.29%	87.41%
	Recall	83.00%	91.67%	87.50%
	Precision	92.22%	97.06%	87.50%
	F1-Score	87.37%	94.29%	87.50%
MobileNetV3 Large	Accuracy	86.00%	95.71%	88.52%
	Recall	85.00%	94.44%	91.91%
	Precision	86.73%	97.14%	86.21%
	F1-Score	85.86%	95.77%	88.97%

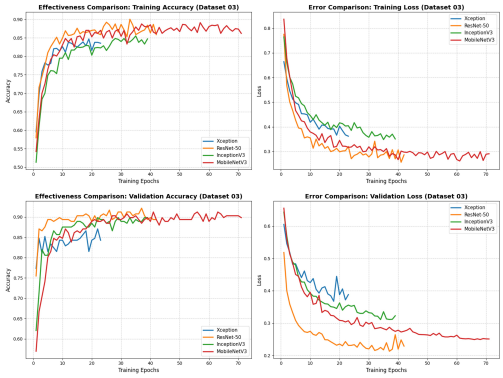


Fig. 5. Comparative learning curves (training and validation accuracy and loss) of the evaluated CNN architectures on Dataset 03.

and achieved the best performance across the metrics when the dataset has clear visual patterns. It also achieved the best overall accuracy on Dataset 03, with InceptionV3 at 88.89%. In Dataset 01, the architecture achieved acceptable accuracy (86.50%) but remained in third place. However, given the challenge of avoiding false negatives (a "bad" tomato classified as "good" would result in the end customer receiving a spoiled or damaged tomato), recall must be considered when selecting an architecture. In this metric, InceptionV3 achieved the best results only on Dataset 02 (tying with ResNet-50), reinforcing the previously noted behavior of performing better when classes are clearly separable.

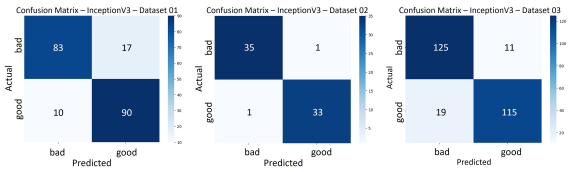


Fig. 6. Confusion matrices for InceptionV3.

The ResNet-50 architecture achieved the highest recall among the architectures presented, ranging from 89% in Dataset 01 to 97.22% in Dataset 02, and reaching 92.65% in Dataset 03, as noticed in Table V, indicating its greater viability when the challenge requires the correct classification of defective tomatoes (considered as "positive" in this study). Therefore, it is the most promising model for the table tomato grading problem based on the studied metrics. Analyzing the learning curves in Figs. 3 to 5, it is noted that the model achieves high validation rates within a few epochs.

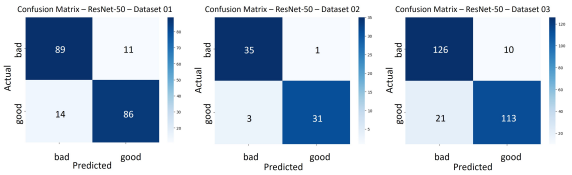


Fig. 7. Confusion matrices for ResNet-50.

In contrast, the Xception network prioritized precision, avoiding false positives. This behavior is recommended when, for the grading problem, it's more important to avoid classifying a good tomato as bad. This model achieved the highest precision on Dataset 01 (92.22%), among all models evaluated on this dataset. However, this model showed slightly lower recall than the other models analyzed, which limits its viability for the proposed problem, given the greater need to avoid false negatives. In the validation curves (Figs. 3 to 5), Xception struggled to maintain long-term generalization compared to the other structures, triggering early stopping mechanisms earlier.

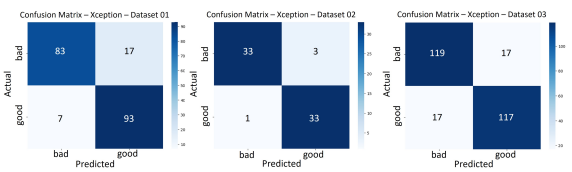


Fig. 8. Confusion matrices for Xception.

MobileNetV3 Large, in turn, achieved the worst performance among the architectures. It is also worth noting that

it took more training epochs to achieve its best performance (Figs. 3 to 5). Although it did not achieve the best performance, MobileNets shouldn't be discarded at this stage of the study, given their architecture, which is ideal for implementation on an embedded system that may be necessary for the problem at hand. Therefore, it's important to consider other metrics, such as classification time and computational cost, to determine the best architecture for industrial implementation.

The confusion matrices for all experiments are shown in Figs. 6 to 9 for qualitative analysis.

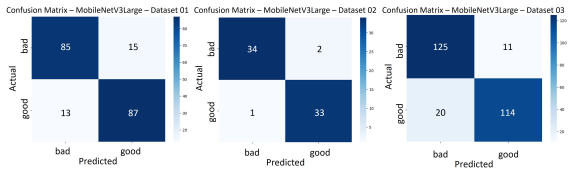


Fig. 9. Confusion matrices for MobileNetV3 Large.

VI. Limitations

This study aimed to compare accuracy, recall, precision, and F1-Score. For this kind of problem, it's also relevant to compare computational cost and object-classification time. This matters because industrial applications need low latency, often best met with embedded systems. Also, processing units may be in rural areas, limiting cloud use. Therefore, computational cost and inference time must be measured to ensure that the model can be deployed at scale.

Another important limitation is the size of the datasets used. Although it was compared against three datasets, it is important to test on larger datasets to improve the models' generalization capacity and reduce difficulties caused by specular shine. These topics will be addressed in future work.

VII. Conclusion and Future Work

Deep learning proved promising for table tomato grading, with all proposed models achieving accuracy of 86% or higher and recall ranging from 83% to 97.22% across all datasets, as demonstrated in Table V. Among the models studied, ResNet-50 proved most promising, achieving recalls of 89%, 97.22%, and 92.65% on Datasets 01, 02, and 03, respectively, as shown in Table V. We also noted that using more images with different defect types improves generalization.

Future work will expand dataset size to evaluate effects on metrics and measure computational cost and classification speed. It will also test additional ResNet variants (ResNet-50V2, ResNet-101, ResNet-152) to assess performance for this task.

Since prior work reports strong results for hybrid models [6] [7], we will develop and test a hybrid model that maximizes accuracy, recall, and precision while minimizing computational cost and inference time. We will compare it with existing architectures to identify the best option for real-world use and to advance Brazilian tomato-grading by integrating hardware and software.

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